

KAYRA ŞENER

Data Scientist | Data Analyst | Statistical Modeling & Simulation | Optimization and ML
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EDUCATION

Boğaziçi University, B.S. in Industrial Engineering

Istanbul, Turkey

GPA: 3.33/4.00 (Cumulative)

Graduated June 2025

Thesis: Designing a Pattern-Based Clustering Approach for Dynamic Simulation Output, applying unsupervised ML and pattern recognition to large simulation datasets

Relevant Coursework: Nonlinear Models in Operations Research, Stochastic Models in Operations Research, Statistical Forecasting & Time Series Analysis, Risk Assessment, Graph Algorithms & Applications, Agent-Based Modeling & Simulation, Systems Simulation, Quality Engineering, Probability & Statistics for Industrial Engineers, Differential Equations, Matrix Theory, Money & Banking, Financial & Cost Accounting, Engineering Economics, Macroeconomics, Microeconomics

Certificates: Understanding ChatGPT, Web Scraping with Python, SQL (Introductory & Intermediate), Effective Speaking

PROJECTS

Dynamic Simulation Behavior Clustering

Undergraduate Thesis, GitHub

- Built a Python pipeline that converts raw multivariate simulation trajectories into interpretable behavioral feature vectors (linearity, oscillation, growth/decay signatures) via OLS regression and detrended peak analysis, then applies hierarchical clustering with optional K-Means sub-clustering to sort simulation runs into a labeled behavioral taxonomy, evaluated against ground truth using Adjusted Rand Index and Adjusted Mutual Information.

Coffee Quant: Regime-Aware Probabilistic Forecasting

Personal Project, GitHub

- Designed an end-to-end quantitative research pipeline for coffee futures: screened 28 exogenous macro, FX, climate, and commodity variables via four independent statistical tests (Granger causality, Spearman cross-correlation, LASSO, Random Forest importance), fed the survivors into a competing model suite (HMM regime detection, GARCH, SARIMAX vs. ARIMA control, Bayesian regression, ensembles), and validated every result with walk-forward backtesting, Brier skill scores, and permutation significance testing rather than a single train/test split.

Agent-Based Modeling: Large-Scale Factorial Simulation Study

Course Project, GitHub

- Built a NetLogo ABM with a fully specified conceptual model and formal verification/validation procedures; ran a large factorial simulation study (48 policy/scenario combinations x 100 replications) and analyzed results with heatmaps and two-sample hypothesis testing to isolate statistically significant parameter effects.

...and several more data, forecasting, and simulation projects on [GitHub](#).

PROFESSIONAL EXPERIENCE

TFM

Kampala, Uganda / Istanbul, Turkey

Data / Operations Specialist

September 2024 – Present

- Managed and analyzed operational data across production, quality, inventory, and sales, using Python and Excel to identify defect patterns, bottlenecks, and process inefficiencies; applied statistical process control and root-cause analysis to support improvements in defect and return rates and production throughput.
- Built and maintained KPI dashboards, forecasting models, and operational systems to track production, inventory, sales, and project performance; developed a database-backed product catalog and order-management system to centralize information, improve coordination, and support demand forecasting, inventory planning, pricing, and working-capital decisions across a multi-country retail and wholesale operation.
- Coordinated day-to-day operations across production, procurement, inventory, sales, and export logistics, working with a 25+ person production workforce and international supplier network to maintain schedules, resolve bottlenecks, and support on-time delivery.
- Closed high-value client deals and directed 12 luxury residential design projects. Used AutoCAD, SketchUp, D5 Render, and Adobe Photoshop for designing.

IDA Process Control Automation

Istanbul, Turkey

IT Intern

July 2024 - September 2024

- Automated a daily sensor data pipeline (Bash to SQL to Python) and developed a custom peak/valley detection algorithm to extract valve-opening events from noisy time-series signals, cross-checking timestamps against expected schedules to flag anomalies; separately designed a SCADA/HMI monitoring system for a chemical factory's tanks and piping network wired to live sensor values.

TradeTurk

Istanbul, Turkey

Operations Part Timer

February 2021 - June 2024

- Managed full-cycle international trade operations, including supplier vetting, contract negotiation, freight forwarding, customs clearance, and last-mile delivery, for Turkish exports to 30+ countries, maintaining relationships with a portfolio of repeat international buyers.
- Performed data analysis on shipment, cost, and supplier records to model commodity pricing and margins, informing the launch and negotiation of supplier contracts for an independent commodities trading branch (coffee, sugar) targeting 450+ container shipments by end-2026.
- Led development of a live B2B data platform with real-time inventory tracking and automated order workflows; tracked FX exposure and landed-cost variance in Excel, building recurring KPI reports (on-time delivery, cost per shipment, order cycle time) for stakeholder review.
- Sourced and closed new client and supplier partnerships through direct sales outreach and negotiation across Europe, the Middle East, and Africa, organizing cross-border logistics across freight, customs, and delivery teams while managing payment terms and contract risk.

TECHNICAL SKILLS

Core Languages (Primary): Python (NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, BeautifulSoup/Scrapy), SQL (PostgreSQL, MySQL), R/RStudio, Excel (Advanced: pivot tables, statistical functions, dashboarding)

Data Science & Machine Learning: Supervised & Unsupervised Learning, Clustering, Regression & Classification, Feature Engineering, Statistical Forecasting & Time Series Analysis, Hypothesis Testing, A/B Testing, Exploratory Data Analysis (EDA), Data Wrangling & ETL, Pattern Recognition, Data Mining, Model Evaluation & Validation, NLP

Simulation & Stochastic Modeling (Primary): Discrete-Event Simulation (Arena), System Dynamics / Stock-and-Flow Modeling (Vensim), Agent-Based Modeling (NetLogo), Monte Carlo Simulation, Stochastic Modeling, Scenario Analysis, Regime Detection

Optimization & Quantitative Methods: Linear & Integer Programming (LP, MIP), Nonlinear Optimization, Stochastic Optimization, Graph Algorithms, Network Flow, Heuristics & Metaheuristics, Risk Assessment & Modeling, Statistical Process Control, Bayesian Methods

Data Engineering: Spark, Hadoop, Kafka, Hive, Relational Database Design & CRUD, Data Pipeline Concepts (ingestion, structuring, storage), REST API, Git Version Control

AI, LLMs & Intelligent Systems: LLMs, Generative AI, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Agents & Agentic Workflows, Tool Use, Embeddings & Vector Search, Semantic Search, Structured Output, LLM Evaluation, OpenAI - Claude API, Hugging Face Transformers, LangChain, LangGraph, FAISS, ChromaDB, Ollama

Secondary Languages & Tools: JavaScript, HTML/CSS, PHP, PowerShell, Bash

Domain Knowledge: Supply Chain Management, Lean Manufacturing, International Trade, ERP/MRP Concepts, Product Lifecycle Management, GDPR / KVKK Privacy Law, Financial Modeling & Accounting, Risk Assessment, Banking & Financial Markets, Investment & Engineering Economics, Macroeconomic Analysis

Hobbies: Playing Drums and Electric Guitar, Chess, Photography, Running